

# KOWSHIK DAS USHNA

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 LinkedIn |  Google Scholar |  GitHub

## EDUCATION

### • Bachelor of Science in Computer Science and Engineering

March 2022 - August 2026

Chittagong University of Engineering and Technology (CUET)

Chittagong, Bangladesh

CGPA - 3.31 (average 3.68 in last four semesters)

### • Higher Secondary

2018 - 2020

MC College

Sylhet, Bangladesh

GPA - 5.00

## RESEARCH INTEREST

• Artificial Intelligence • Machine Learning • LLM • NLP • Computer Vision

## SKILLS

- **Programming Languages:** Python, C++, HTML, CSS
- **ML/DL Frameworks:** TensorFlow, PyTorch, Scikit-learn, HuggingFace, LangChain, FAISS
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, OpenCV
- **Tools & Technologies:** Git, GitHub, Flask, Streamlit, Overleaf, MS Office

## NOTABLE PROJECTS

- RAG-Based Multi-Document Question Answering System using LangChain, FAISS and HuggingFace Models
- CNN-Based Student Attentiveness Detection with Dimensionality Reduction and High-Accuracy Classification
- AI-Powered JSON Generator for Structured Output Automation using LLM-Based Chatbot Framework
- Hybrid CNN-LSTM Architecture for Large-Scale Tweet Sentiment Analysis
- DeepLabV3-Based Bokeh Effect Image Processing System with Real-Time Web Interface

## THESIS

### 1. *A Multi-Partition Retrieval-Augmented Generation Based Approach for Improving Large Language Model Performance.*

- Proposed a training-free alternative to reinforcement learning-based RAG by integrating fuzzy-match retrieval and task-specific decoding.
- Supervised by Md. Rashadur Rahman, Assistant Professor, CSE, CUET.

## PUBLICATIONS

### Conference Papers

1. Nazmus Sakib Md Adil, **Kowshik Das Ushna**, Md. Tasnimur Rahman, Shuhena Salam Aonty, *CNN-Based Student Attentiveness Detection: A Hybrid Approach with Dimensional Reduction*. In 8th International Conference on Engineering Research, Innovation and Education 2025 (ICERIE 2025). Atlantis Press (Part of Springer Nature). DOI: [10.2991/978-94-6463-884-4\\_81](https://doi.org/10.2991/978-94-6463-884-4_81)
2. Farhan Amin, Md. Tasnimur Rahman, **Kowshik Das Ushna**, Nazmus Sakib Md Adil, Shuhena Salam Aonty, *Earthquake Magnitude Prediction Using Feature-Engineered Machine Learning Models in the South Asia Region*. QPAIN 2026. DOI: [10.1109/QPAIN69676.2026.11545587](https://doi.org/10.1109/QPAIN69676.2026.11545587)

### Journal Articles

1. Md. Rakib Hossain, Shuhena Salam Aonty, Mujtahid Alam, **Kowshik Das Ushna**, *AXQ-UNet: Attention-based Explainable Quantum U-Net Framework for Brain Tumor Segmentation and Classification*. Under review at Springer Nature.
2. Farhan Amin, **Kowshik Das Ushna**, Md. Tasnimur Rahman, Nazmus Sakib Md Adil, Shuhena Salam Aonty, *Behavioral Feature-Enriched Customer Segmentation and Strategy Prescription in E-Commerce: A Hybrid Machine Learning Approach*. Under review at Elsevier.

## EXPERIENCES

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◦ **Industrial Attachment Trainee** | The Tork

Nov 3 – Nov 16

- \* Completed a two-week industrial attachment, gaining practical knowledge of software development workflows, system architecture and teamwork.
- \* Developed an AI-powered JSON Generator using a chatbot framework to produce structured outputs for workflow automation.
- \* Collaborated with senior engineers, participated in code reviews and applied Agile development practices.

## LEADERSHIP AND ACTIVITIES

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- **Vice President**, Sylhet Students' Forum, CUET
- **Vice President**, Public University Students Association of Sunamganj
- **Organizing Secretary** Andromeda Space & Robotics Research Organization Research (ASRRO).
- **Joint General Secretary**, IEEE CS Student Branch.

## REFERENCES

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**1. Md. Rashadur Rahman**

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Chittagong University of Engineering & Technology (CUET)  
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